

Lilly Chiavetta

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EDUCATION

Duke University, Durham, NC

Dec 2025

Master of Engineering in Electrical & Computer Engineering

Bachelor of Science in Electrical & Computer Engineering and Biomedical Engineering

GPA: 3.9/4.0

- Relevant Coursework: Robotics, Engineering Design & Communication, Data Structures and Algorithms, Microelectronics, Signals and Systems, Embedded Systems, Medical Device Design, Computer Architecture
- Honors & Awards: Chief Student Marshal (2024), Outstanding Teaching Assistant (2024, 2025), 1st place in HackDuke (2022 & 2023), Best Overall Robot Design (ECsE Fundamentals- 2022)

PROFESSIONAL EXPERIENCE

Stryker, *Electrical Engineering Intern*, Weston, FL

May 2025-Aug 2025

- Designed requirements, schematic, and 8-layer PCB for a power tool controller that passed external review.
- Developed automated testing scripts with a medical robotic system and used python to analyze test results.
- Completed concept development and initial proof of concept models for a new robot arm calibration tool.

Renesas Electronics, *Systems Engineering Intern*, Morrisville, NC

May 2024-Aug 2024

- Designed a smart damper control circuit and algorithm in C with environment sensors and motor control
- Designed and simulated a schematic and PCB in Altium for a TRIAC motor control embedded system

Renesas Electronics, *Applications Engineering Intern*, Morrisville, NC

May 2023-Aug 2023

- Conducted testing on various MCUs paired with diverse sensor and motor configurations using C.
- Started the creation of reusable blocks in Altium Designer, contributing to the company's first fully functional set and delivered a comprehensive instructional presentation to an international set of teams

Duke Engage, *Engineering Fellow*, Kampala, Uganda

May 2022-July 2022

- Lived with an international team in Uganda and worked to design and build a cost effective fluid warmer
- Successfully designed and prototyped an embedded system meant to maintain temperature with accuracy

Duke University Pratt School of Engineering, *Teaching Assistant*, Durham, NC

Jan 2023-Present

- Effectively explain and demonstrate electrical and computer engineering concepts to students
- Lead a lab focused on Microelectronics, explaining niche concepts and how to use lab bench tools
- Assisted the instruction of ECE110: Fundamentals of ECE and ECE230: Introduction to Microelectronics

LEADERSHIP/EXTRACURRICULAR EXPERIENCE

Duke Robotics Club, *President* (2022-2024), Durham, NC

Aug 2021-Present

- Manage three sub-teams in the production of a fully functional AUV utilizing ROS, complete with computer vision and acoustics sub-systems; ensuring a cohesive design that fits all the robot's functionality needs
- Designed PCBs for active filtering of hydrophone signals and an algorithm localizing an underwater pinger
- Co-Authoring a technical report that received first place at the International Robosub competition

HackDuke, *First Place Recipient*, Durham, NC

Sept 2023

- Used C++ and Python in a device used to remotely watch many patients to reduce nurses' workload
- Developed a modular system interfacing with multiple sensors to keep track of different patient vitals

TECHNICAL SKILLS

Embedded Systems, Altium Designer, Python, C, C++, Java, ROS2, MATLAB, Circuit Design, PCB, Electronics Lab Equipment, Raspberry Pi, OpenCV, Sensor Integration, Medical Requirements, 3D printing, Project Management